

Akula Chetan Krishna Sai

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Education

B.Tech, Computer Science and Engineering

Nov 2022 – May 2026

VNR Vignana Jyothi Institute of Engineering and Technology, Hyderabad

GPA: 8.12

Experience

Software Engineer Intern – Tower Cloud

June 2026 – Present

- Developed TowerWatch across Go services and a React/Next.js portal, delivering resource overviews, historical graphs, and live monitoring for VMs, databases, storage, load balancers, Kubernetes, and Function Apps.
- Built organization-scoped alert rules, recipient validation, severity settings, and history; integrated Alertmanager webhooks with PostgreSQL-backed transitions and asynchronous email delivery.
- Optimized VictoriaMetrics retrieval with batched queries, bounded caching, and incremental snapshots; handled missing telemetry and scaled-to-zero functions without misleading metrics.
- Extended Helm charts, recording rules, and exporters for worker readiness, pod capacity, and database telemetry; fixed PMM authentication, scrape discovery, and network policies across PostgreSQL, MySQL, and MongoDB.
- Added authenticated capacity checks to provisioning, start, restore, and upgrade workflows across cloud services; implemented correlation-based notification deduplication for retries and concurrent requests.
- Diagnosed Kubernetes deployment and monitoring failures through pod logs, scrape targets, and live metric queries; validated fixes with automated tests and performed scoped Helm rollouts with preflight checks and rollback protection.
- Implemented Nexus registry throughput metrics from access logs, using stateful parsing to preserve counters across log rotation and prevent double-counting chunked uploads.

Projects

SimManCee – Business Simulation Platform

MERN, AWS

- Developed a full-stack business simulation platform using React, Node.js, and REST APIs to connect interactive simulation views with backend services.
- Built responsive, reusable interface modules for simulations and team content, maintaining consistent layouts across different screen sizes.
- Deployed the frontend on AWS S3 and backend on EC2; configured Route 53 for custom-domain access at sim-man-cee.com.

Post-HCT Survival Prediction

Python, Ensemble Learning

- Built a five-model ensemble using LightGBM, two XGBoost variants, HistGradientBoosting, and Random Forest, with Kaplan–Meier target transformation for censored outcomes.
- Achieved a 0.6853 C-index and 0.1930 RMSE using 10-fold cross-validation; evaluated feature importance and performance across six demographic groups.
- Analyzed 28,800 records with 59 clinical and donor-recipient features; assessed multicollinearity and subgroup performance, reporting a race-group C-index standard deviation of 0.0088.

Student Record Management System

Node.js, MongoDB, Docker, Jenkins

- Built secure Node.js/MongoDB REST APIs for student-record CRUD operations, with automated API tests and health checks.
- Containerized the application using Docker and automated build and deployment workflows through Jenkins CI/CD.

Technical Skills

Languages: Go, Python, JavaScript, TypeScript, Java, C++, C, R

Web: React.js, Next.js, Node.js, REST APIs, HTML, CSS

Databases & Messaging: PostgreSQL, MongoDB, SQL, RabbitMQ

Cloud & DevOps: AWS, Kubernetes, Helm, Docker, Jenkins, GitHub Actions, Git

Observability: VictoriaMetrics, Prometheus, Grafana, Loki, Alertmanager, PMM

Core CS: DSA, OOP, DBMS, Operating Systems, Computer Networks

Publications & Patent

- **Research paper:** Co-authored and presented “Post-Hematopoietic Cell Transplantation Survival Predictions using Ensemble Machine Learning Models,” ICIRCA 2026, June 3–5, 2026, pp. 1605–1611.
- **Published patent application:** Co-inventor, “Hybrid BLE–LoRa Infrastructure-Independent Communication System for Long-Range IoT Messaging Without Internet or Gateways.” **IN202641062255 A1**, May 29, 2026.

Achievements & Leadership

TuringCup-2023 finalist; CSI technical event organizer and volunteer; Dramatrix Event Coordinator; regional U-16 badminton bronze medalist; led school football team to U-15 CBSE Clusters semifinals.